

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `53f9440d-42cc-4d77-ab9a-be8ebb36efed`
- **Timestamp:** 2026-09-05T15:14:13.862036Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** The interlocutory applications have been filed seeking direction for payment of rent and other associated b
- **Answer:** I do not have enough information to answer this.
- **Health Score:** 0.07
- **Primary Issue:** RETRIEVAL_MISS
- **Claims Supported:** 0 / 1
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
--- ---	---	---	---
CORPUS	UNKNOWN	0.5	Corpus relevance is inconclusive.
RETRIEVER	FAIL	0.85	No retrieved chunks provide support, and retrieval scores strongly indicate a lack of useful ev
GENERATOR	UNKNOWN	0.5	1 generated claims are unsupported, but retrieval quality is ambiguous.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

Efficiency: 0/6 chunks used (0%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 1 (Supported: 0, Partial: 0, Unsupported: 1, Contradicted: 0)
- **Grounding Score:** 0.00
- **Evidence Coverage:** 0.17
- **Average Entailment:** 0.11
- **Latencies:** Retrieval (3710.7ms) | Generation (7316.9ms) | Verification (8259.4ms)

5. RAGAS Metrics

- **Faithfulness:** 0.0000
- **Answer Relevancy:** 0.3021
- **Context Precision:** 0.0000
- **Context Relevancy:** 0.0000
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[b56b291a-986e-448d-995d-dcea39c746fb]** (UNSUPPORTED): I do not have enough information to answer this.
- *Evidence (Chunk cf11f1b5-7e7c-4d63-9b98-b7d99033a7d4):* 28

MAGISTRATE'S ORDER PROHIBITING THE DOING OF ANYTHING ON LAND OR WATER
(See section 166)

A dispute having arisen concerning the right of use of.....(state
concisely the subject of dispute) situate within my local jurisdiction, the possession of which
land (or water) is claimed exclusively by.....(describe the person or
persons), and it appears to me, on due inquiry into the same, that the said land (or water) has
been open to the enjoyment of such use by the public (or if by an individual or a class of
persons, describe him or them) and (if the use can be enjoyed throughout the year) that the
said use has been enjoyed within three months of the institution of the said inquiry (or if the
use is enjoyable only at a particular season, say, "during the last of the seasons at which the
same is capable of being enjoyed");
I do order that the said.....(the claimant or claimants of
possession) or any one in their interest, shall not take (or retain) possession of the said land

(or water) to the exclusion of the enjoyment of the right of use aforesaid, until he (or they) shall obtain the decree or order of a competent Court adjudging him (or them) to be entitled to exclusive possession.

7. Root Cause Analysis

- **Primary Cause:** RETRIEVAL_MISS
- **Confidence:** 0.88
- **Reasoning:**
 - Starting root cause analysis for trace 53f9440d-42cc-4d77-ab9a-be8ebb36efed.
 - Skipping CORPUS: status is UNKNOWN (insufficient evidence).
 - Failure at RETRIEVER: RETRIEVAL_MISS (confidence=0.85). Observation: No retrieved chunks provide support, and retrieval failed.
 - Skipping CHUNKING: no observable state in matrix.
 - Skipping GENERATOR: status is UNKNOWN (insufficient evidence).
 - GROUNDING passed (confidence=0.90).
 - Primary root cause: RETRIEVAL_MISS at RETRIEVER (earliest upstream failure, confidence=0.85).
 - Analysis complete. Primary cause: RETRIEVAL_MISS.

8. Corrective Action Plan

- **[IMMEDIATE] Increase Retrieval Top-K**
 - Description: Increase the number of retrieved documents before providing context to the LLM.
 - Expected Improvement: Higher recall; missing context is caught within the expanded window. (Metric: Verification support)
- **[SHORT_TERM] Implement Hybrid Search (Dense + Sparse)**
 - Description: Add BM25 lexical search and combine with semantic vector search using Reciprocal Rank Fusion.
 - Expected Improvement: Near-perfect recall for exact-match terminology. (Metric: Retrieval hit rate for exact entity queries)
- **[EXPERIMENTAL] Deploy Cross-Encoder Reranker**
 - Description: Pass the Top-N results through a Cross-Encoder to re-score query-document relevance.
 - Expected Improvement: Drastically improved precision at Top-1 and Top-3. (Metric: Precision@3 increases by at least 20%)
- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
 - Description: Retrieval efficiency is low (0%); consider reducing top-k or tightening the reranker score cutoff.
 - Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- **Primary Cause:** RETRIEVAL_MISS
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** RETRIEVAL_MISS
- **Supporting Claims:** b56b291a-986e-448d-995d-dcea39c746fb
- **Supporting Chunks:** cf11f1b5-7e7c-4d63-9b98-b7d99033a7d4

10. Overall Assessment

- **Major Strength:** GROUNDING is healthy (confidence=90%): No claims are contradicted by retrieved evidence.
- **Major Weakness:** RETRIEVAL_MISS
- **Next Priority:** Increase Retrieval Top-K
- **Recommendation:** Pipeline requires architectural adjustments across multiple stages.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace
artifacts/rag_traces/53f9440d-42cc-4d77-ab9a-be8ebb36efed.json

✓ ClaimSet
artifacts/claim_sets/53f9440d-42cc-4d77-ab9a-be8ebb36efed.json

✓ VerificationResults
artifacts/verification/53f9440d-42cc-4d77-ab9a-be8ebb36efed.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/53f9440d-42cc-4d77-ab9a-be8ebb36efed.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/53f9440d-42cc-4d77-ab9a-be8ebb36efed.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/53f9440d-42cc-4d77-ab9a-be8ebb36efed.json

✓ DiagnosticEvaluationReport

artifacts/reports/53f9440d-42cc-4d77-ab9a-be8ebb36efed.json